AIAP® Batch 25 Technical Assessment

Deadline**: <u>1900 hrs, 31st August 2026</u>**

<u>Tasks & Deliverables</u>

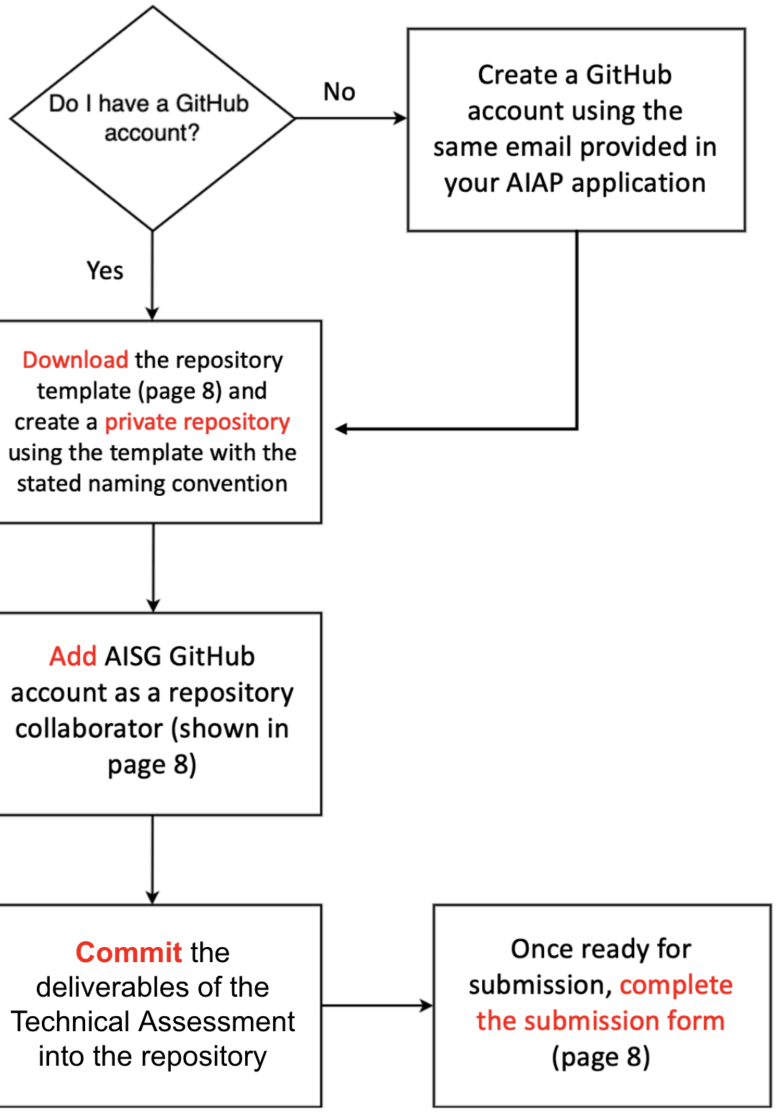
This assessment consists of the following parts:

1. Exploratory Data Analysis in Jupyter Notebook **2. Decision Log** **3. End-to-end Machine Learning Pipeline in Python Scripts (`.py`)** **4. AI Assistant Chat History**

NOTE: Please read the whole document.

Technical Assessment Overview

There are **four** parts to the Technical Assessment: Decision Log, Exploratory Data Analysis, End-to-end Machine Learning Pipeline, and AI Assistant Chat History. You are to attempt all parts and submit the deliverables by uploading them to your own **private** GitHub repository. The following flowchart outlines the major steps for the Technical Assessment. Details will be provided in the subsequent sections of this document.



Problem Statement

Business Context

Travelers' Inn operates two hotels in Singapore: the City Hotel (downtown, in the Kallang area near the National Stadium) and the Airport Hotel (in the far east, near Changi). Between them they take in a high volume of reservations year-round, across a mix of leisure travellers, corporate guests, tour groups, and online-travel-agent bookings.

Over the past year, the group's profit margins have come under pressure. Rooms sit empty more often than management would like — sometimes because a confirmed booking is called off at the last minute, sometimes because a guest simply never shows up — and the rates charged for the rooms that to sell are set largely by intuition. The General Manager suspects the group is leaving significant money on the table every month, but is not sure where the biggest opportunity lies.

The Head of Revenue & Reservations called your team in. In the conversation, she said:

> "*Our profit margins just aren't where they need to be, and I'm convinced there's money being left on the table somewhere. We'd like to use AI to help. Can your team build us something?*"
>
> — Head of Revenue & Reservations, *Travelers' Inn*

After some discussion on this, she added:

> "*Frankly, there are several areas I know that are hurting us. We lose a lot of rooms at the last minute due to fall through or no-shows and we rarely see it coming. Our room rates are mostly gut feel and fixed, we get mixed feedback from customers, sometimes saying we are very cheap and/or very expensive. It feels like we are under-pricing on some nights and overpricing on others. Find me the best angle you possibly can. Lastly, expansion of rooms and staff is out of the question.*"
>
> — Head of Revenue & Reservations, *Travelers' Inn*

You have access to 3 years of Travelers' Inn booking records and a set of post-stay guest reviews.

Objectives

Your task is to:

1. Refine these vague requests into a well-defined problem statement that you will work on. Document your thought process in the Decision Log.
2. Perform an Exploratory Data Analysis on the dataset provided.
3. Build an end-to-end machine learning pipeline that addresses the problem statement you defined.
4. Document your use of AI code assistant(s) by submitting a Prompt Chat History alongside your work.

Dataset

You can query the datasets from:
https://techassessment.blob.core.windows.net/aiap25-assessment-data/hotel.db

The dataset can be accessed through the `hotel.db`. You may find either of the following packages, `SQLite` or `SQLAlchemy`, useful for accessing this database.

You should place the `hotel.db` file in a `data` folder. Your machine learning pipeline should retrieve the dataset using the relative path `hotel.db`.

DO NOT upload the `hotel.db` onto your GitHub repository.

Table 1 - List of Attributes

Attribute	Description
booking_id	Unique reservation identifier.
client_email	Guest's registered email address. Primary key for reviews
hotel	Which property the reservation is for: City Hotel or Airport Hotel.
is_canceled	Whether the reservation was cancelled (1) or honoured (0).
lead_time	Number of days between the date the booking was made and the arrival date.
arrival_date_year	Year of the arrival date (2023–2025).
arrival_date_month	Month of the arrival date
arrival_date_week_number	ISO week number of the arrival date (1–53).
arrival_date_day_of_month	Day of the month of the arrival date (1–31).
stays_in_weekend_nights	Number of weekend nights (Saturday/Sunday) booked.
stays_in_week_nights	Number of week nights (Monday–Friday) booked.
adults	Number of adults on the reservation.
children	Number of children on the reservation.
babies	Number of babies on the reservation.

meal	Meal package booked: BB (Bed & Breakfast), HB (Half Board), FB (Full Board), SC (Self Catering), Undefined (no package).
country	Country of origin of the guest, as an ISO country code (e.g. SGP, MYS, IDN).
market_segment	Market segment of the booking: Online TA, Offline TA/TO, Direct, Groups, Corporate, Complementary, Aviation, Undefined.
distribution_channel	Booking distribution channel: TA/TO, Direct, Corporate, GDS (Global Distribution System)
is_repeated_guest	Whether the guest was a returning guest at booking time (1) or first-time (0).
previous_cancellations	Number of prior bookings by the same customer that were cancelled before this reservation.
previous_bookings_not_canceled	Number of prior bookings by the same customer that were not cancelled before this reservation.
reserved_room_type	Code of the room type the guest reserved (anonymised as single letters A–L).
assigned_room_type	Code of the room type actually assigned at check-in. May differ from the reserved type (e.g. overbooking, upgrade, guest request).
booking_changes	Number of amendments made to the booking between creation and check-in / cancellation.
deposit_type	Deposit arrangement: No Deposit, Non Refund, Refundable.
agent	Identifier of the travel agency that made the booking. Missing when no agency was involved.
company	Identifier of the company / entity responsible for payment. Missing for most bookings.
days_in_waiting_list	Number of days the booking was on the waiting list before it was confirmed.
customer_type	Type of booking: Transient, Transient-Party, Contract, Group.
adr	Average Daily Rate, in SGD — total lodging revenue for the stay divided by the total number of nights.

required_car_parking_spaces	Number of car parking spaces requested by the guest.
total_of_special_requests	Number of special requests made by the guest (e.g. high floor, twin beds, late check-out).
reservation_status	Most recent status: Check-Out, Canceled, or No-Show.
reservation_status_date	Date on which the last reservation_status was set (YYYY-MM-DD).
promo_code	Various types of promo code used

Table 2 - List of Attributes

Attribute	Description
client_email	Guest's registered email address. Foreign key linking back to bookings.client_email.
rating	Overall guest satisfaction rating, from 1 (very dissatisfied) to 5 (very satisfied).
comments	Free-text comment left by the guest.

Submission Requirements — Four Deliverables

Task 1: Exploratory Data Analysis (EDA)

Using the dataset specified in the **Dataset** section, conduct an EDA and create an interactive notebook (.ipynb file) in **<u>Python</u>** that can be used as a presentation to explain the findings of your analysis. It should contain appropriate visualisations and explanations to assist readers in understanding how these elaborations are arrived at and their implications.

Deliverable

- Jupyter Notebook in **<u>Python</u>**: a `.ipynb` file named **`eda.ipynb`**. (do adhere to the naming requirement)

Evaluation

In the submitted notebook, you are required to

- Outline the steps taken in the EDA process
- Explain the purpose of each step
- Explain the conclusions drawn from each step
- Explain the interpretation of the various statistics generated and how they impact your analysis
- Generate clear, meaningful, and understandable visualizations that support your findings
- Organise the notebook so that is it clear and easy to understand

Please note that your submission will be heavily penalised for any of the following conditions:

- `.ipynb` missing in the submitted repository
- `.ipynb` cannot be opened on Jupyter Notebook
- Explanations missing or unclear in the submitted Jupyter Notebook

Task 2: Decision Log

Complete and submit the `decision_log.md` file found in the template repository folder (download link below). The template contains five fixed questions:

- Q1 — Clarifying questions
- Q2 — Defining the Problem Statement
- Q3 — Key decisions during the solution development
- Q4 — Use of the AI assistant
- Q5 — Next steps

Deliverable

- `decision_log.md`

Evaluation

This decision log is the primary instrument by which we understand your thinking. The questions below cover the reasoning behind your work — from how you scoped the problem at the start to the decisions you made during the work itself. Do answer all five questions in your own words. You may use AI assistance freely on the technical deliverables (the EDA and the ML pipeline), but this Decision Log itself should be written by you — it is the record of your own thinking that we cross-check against your chat history.

Task 3: End-to-end Machine Learning Pipeline

Design and create a machine learning pipeline (MLP) in Python scripts (`.py` files) that will ingest and process the entailed dataset, subsequently, feeding it into the machine learning algorithm(s) of your choice.

Do not develop your MLP in an interactive notebook.

The pipeline should be easily configurable to enable easy experimentation of different algorithms and parameters as well as ways of processing data. You can consider the usage of a config file, environment variables, or command line parameters.

Within the pipeline, data must be fetched/imported using SQLite, or any similar packages.

Deliverables

1. A folder named `**src**` containing Python modules/classes in `.py` format.
2. An executable bash script `**run.sh**` at the base folder of your submission to run the aforementioned modules/classes/scripts. DO NOT install your dependencies in the `run.sh`; this will be taken care of automatically when we assess the assignment if you have created your `requirements.txt` correctly.
3. A `**requirements.txt**` file in the base folder of your submission.
4. A `**README.md**` file that sufficiently explains the pipeline design and its usage. You are required to explain the thought process behind your submitted pipeline in the README. The README is expected to contain the following:
    - Full name (as in NRIC) and email address (stated in your application form).
    - Overview of the submitted folder and the folder structure.
    - Instructions for executing the pipeline and modifying any parameters.
    - Description of logical steps/flow of the pipeline. If you find it useful, please feel free to include suitable visualisation aids (eg, flow charts) within the README.
    - Overview of key findings from the EDA conducted in Task 1 and the choices made in the pipeline based on these findings, particularly any feature engineering. Please keep the details of the EDA in the `.ipynb`. The information in the `README.md` should be a quick summary of the details from `.ipynb`.
    - Describe how the features in the dataset are processed (summarised in a table).
    - Explanation of your choice of models for each machine learning task.
    - Evaluation of the models developed. Any metrics used in the evaluation should also be explained.
    - Other considerations for deploying the models developed.

Evaluation

The submitted MLP, including the `README.md`, will be used to assess your understanding of machine learning models/algorithms as well as your ability to design and develop a machine learning pipeline. Specifically, you will be assessed on

- Appropriate data preprocessing and feature engineering
- Appropriate use and optimization of algorithms/models
- Appropriate explanation for the choice of algorithms/models
- Appropriate use of evaluation metrics
- Appropriate explanation for the choice of evaluation metrics
- Understanding of the different components in the machine learning pipeline

In your submitted Python scripts (`.py` files), you will be assessed on the quality of your code in terms of reusability, readability, and self-explanatory.

Please note that your submission will be penalised for any of the following conditions:

- Incorrect format for `requirements.txt`
- `run.sh` fails upon execution
- Poorly structured `README.md`
- Disorganised code that fails to make use of functions and/or classes for reusability
- MLP not submitted in Python scripts (`.py` files), including MLP built using Jupyter Notebooks.

Note for Windows users

DO NOT submit a Windows batch (`*.bat`) script in replacement of the bash script. Use either 'Windows Subsystem for Linux (WSL)' or 'Git Bash'/'cygwin' for the creation of the bash script.

Task 4: AI Assistant Chat History

You are permitted to use AI code assistant(s) throughout this assessment. We do not penalise the use of AI; we assess the quality of your use of it.

To support that, complete and submit the `prompt_chat_history.md` file from the template repository (download link below). Your submission has two parts:

- Share links — one per AI conversation, for every tool that supports a share-link feature (Claude, ChatGPT, Gemini, etc.).
- Raw, unedited transcripts — produce each conversation into the template, labelling messages as [User] or [AI]. Don't edit, summarise, or rewrite anything afterwards.

For tools without share-link support (e.g., GitHub Copilot Chat, Cursor, Claude Code in agent mode), mark the share-link as “not available” and rely on the transcript portion.

Deliverables

- `prompt_chat_history.md`

Evaluation

In your Prompt Chat History, we assess the quality of your AI use — how you direct the assistant and engage critically with its output. The chat is read alongside your Decision Log to check coherence between the two.

Note: If there is too much information, keep it in a folder and describe accordingly in README.

Submission Format

Please complete **ALL** of the following steps

1. Create a GitHub account using the **same** email provided in your AIAP application form.

2. Download the repository template from:
   https://techassessment.blob.core.windows.net/aiap25-assessment-data/aiap25-NAME-NRIC.zip

Note:
The downloaded repository template contains a hidden folder: `.github`. The `.github` folder contains scripts to execute your end-to-end machine learning pipeline using GitHub Actions. Specifically, it will first install the required dependencies using your `requirements.txt` and subsequently, execute your bash script (`run.sh`). You can manually trigger the pipeline under Actions in your repository.

3. Using the downloaded template, create a **private** repository using the following naming convention:

   aiap25-<full name (as in NRIC) separated by dashes>-<last 4 characters of NRIC>

   For example, `*aiap25-john-lim-der-hui-321A*`.

   The default branch should be called 'main'.

4. Add the following account as a collaborator in your private repository:
   - Username: **AISG-AIAP**
   - Email: **aiap-internal@aisingapore.org**

Your repository should adhere to the following structure:

```
aiap25-john-lim-der-hui-321A/
├── README.md
├── eda.ipynb
├── src/
│   └── ...
├── run.sh
├── requirements.txt
├── decision_log.md
└── prompt_chat_history.md
```

README.md	Pipeline overview, run instructions, EDA summary
eda.ipynb	Deliverable 1: EDA notebook
decision_log.md	Deliverable 2: completed template
src/	Deliverable 3: ML pipeline code
run.sh	Deliverable 3: pipeline runner
requirements.txt	Deliverable 3: dependencies
prompt_chat_history.md	Deliverable 4: share links + transcripts

Do not include the dataset files in your submission.

5. We encourage you to adhere to Git best practices. Once your repository is ready for submission, complete the following form at Google Form Link

NOTE: During the assessment period, you are still allowed to make changes to your repository after submitting the form.

CAUTION

1) Failing to complete any task and deliverable will result in penalties to your submission.